PURDUE
UNIVERSITY

Spring 2026, CE 203 Syllabus
Principles and Practice of Geomatics

Course Information

Lecture time: MWF 11:30 AM – 12:20 PM
Room: Brown Hall 170
Lab time: TBD
Course credit hours: 4 credits
Course web page/BrightSpace page: https://purdue.brightspace.com/

Instructors

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Teaching Assistants

Name: Lena Azimi
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Name: Arnav Goel (Lead TA)
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Name: Aser Nader
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Name: Sungwoon Hyung
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Name: Mona Hodaei
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Name: Pramod Parande
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Name: Hansae Kim
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Course Description

Basic surveying measurements and computations for engineering project control, mapping, and construction layout; theory of observational errors and error propagation; fundamental concepts of horizontal and vertical control systems; use of topographic maps and plan-profile sheets; computation of horizontal and vertical curves; introduction to computer tools used in Civil Engineering.

Learning Outcomes

The objectives of this course are that you become familiar with the concepts, capabilities, instrumentation, and operations of Geomatics through lecture presentations and extensive hands-on lab exercises. This includes various modes of geospatial data collection, data analysis, data presentation, and adjustment. In support of those objectives, you will be asked to develop:

- An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics (ABET #1)
- An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgement to draw conclusions (ABET #6)

Learning Resources, Technology, & Texts

- Required Textbook: No textbook is required in this course.
- BrightSpace page
    - You can access the course via BrightSpace. It is strongly suggested that you explore and become familiar not only with the site navigation, but with content and resources available for this course.

Instructor and TA's Email Availability and Policies

Instructors and TAs will be available via email daily and try to respond as soon as possible (generally within 24-48) hours. When emailing, please place the course number/section and the topic in the subject line of the email (*e.g.,* CE203 – Assignment 2 Question). This will help instructors and TAs tremendously in locating and responding to your emails quickly.

Assignments and Points

Your learning will be assessed through a combination of participation, quizzes, lab work, homework assignments, exams (midterm and final) throughout the semester. Details on these assignments and exams, including rubrics to guide evaluation, and guidelines on discussion participation and evaluation will be posted on the course website.

Assignments	Weight (%)
Participation and quizzes	15
Lab work and homework assignment	45
Midterm	20
Final Exam	20

Missed or Late Work

If you must miss a class or lab, send the instructor an email with explanation before the fact. Missed assignments may only be made up when you notify the instructor ahead of time with an explanation and plan for completion. These requests will be accepted at instructors' discretion and may include a point penalty of 10% per day late. Asking for an extension does not guarantee it will be granted. If you miss a lab you may ask permission of a TA to join another lab section for that session.

Grading Scale

In this class grades reflect the sum of your achievement throughout the semester. You will accumulate points as described in the assignments portion above, with each assignment graded according to a rubric. At the end of the semester, final grades will be calculated by weighting the total points earned and translating those numbers (out of 100) into the following letters.

- A+: 95 - 100
- A: 90 – 94.99
- A-: 85 – 89.99
- B+: 80 – 84.99
- B: 75 – 79.99
- B-: 70 – 74.99
- C+: 65 – 69.99
- C: 60 – 64.99
- C-: 55 – 59.99
- D+: 50 – 54.99
- D: 45 – 49.99
- D-: 40 – 44.99
- F: 39.99 or below

Incompletes

A grade of incomplete (I) will be given only in unusual circumstances. To receive an "I" grade, a written request must be submitted at least one week before the end of the course, and approved by the instructor. The request must describe the circumstances, along with a proposed timeline for completing the course work. Submitting a request does not ensure that an incomplete grade will be granted. If granted, you will be required to fill out and sign an "Incomplete Contract" form that will be turned in with the course grades. Any requests made after the course is completed will not be considered for an incomplete grade.

Course Schedule

Subject/Faculty	Week	Lecture	Date	Topic	Lab
Remote Sensing: Photogrammetric Mapping and Laser Scanning **(Habib)**	1	1	1/12/24	Geomatics Engineering: Definition and Applications	Photogrammetric Camera, Visual Perception, Film Development, 3D Reconstruction
		2	1/14/24	Photogrammetric Mapping: Definition and Applications	
		3	1/16/24	Optics, Cameras, and Vertical Photography	
	2	4	1/19/24	**No Class (MLK Day)**	Vertical Photography and Relief Displacement
		5	1/21/24	Optics, Cameras, and Vertical Photography	
		6	1/23/24	Optics, Cameras, and Vertical Photography	
	3	7	1/26/24	Optics, Cameras, and Vertical Photography	Photogrammetric Mission Planning, Relief Displacement, and Parallax Computations
		8	1/28/24	Optics, Cameras, and Vertical Photography	
		9	1/30/24	Photogrammetric Mission Planning	
	4	10	2/02/24	Vector/Matrix Manipulation	Propagation of Errors/Uncertainty
		11	2/04/24	Error/Uncertainty Propagation	
		12	2/06/24	Error/Uncertainty Propagation	
	5	13	2/09/24	Rotation Matrices	Propagation of Errors/Uncertainty & Photogrammetric Mathematical Model
		14	2/11/24	Collinearity Equations and Theory of Orientation	
		15	2/13/24	Collinearity Equations and Theory of Orientation	
	6	16	2/16/24	Georeferencing of Remote Sensing Systems: Indirect Georeferencing	Bundle Adjustment with Self Calibration
		17	2/18/24	Georeferencing of Remote Sensing Systems: Direct Georeferencing	
		18	2/20/24	Laser Scanning: Introduction	
	7	19	2/23/24	Laser Scanning: Operational Principles	Static and Mobile Laser Scanning Systems
		20	2/25/24	Laser Scanning: Mathematical Model	
		21	2/27/24	Laser Scanning: Applications	
Geographic Information System **(Jung)**	8	22	3/02/24	Review	
		23	3/05/24	**Midterm Exam (Remote Sensing: Photogrammetric Mapping and Laser Scanning)** PHYS 112	

		24	3/06/24	Introduction and Concepts	
	9	25	3/09/24	Attribute Data Processing	Introduction to ArcGIS
		26	3/11/24	Maps and Mapping	
		27	3/13/24	Datums and Map Projections	
	10	28	3/16/24	**No Class (Spring Break)**	
		29	3/18/24		
		30	3/20/24		
	11	31	3/23/24	Data Models	GIS Map Generation
		32	3/25/24	National Geographic Data	
		33	3/27/24	Spatial Analysis	
Surveying, Geodesy, GPS **(Jung)**	12	34	3/30/24	Differential Leveling	Elements of Surveying Measurements
		35	4/01/24	Elevation Datums	
		36	4/03/24	Leveling Closure and Adjustment	
	13	37	4/06/24	Angle and Distance Measurements	Leveling and its Computation
		38	4/08/24	Traverse Computations	
		39	4/10/24	Traverse Computations and Adjustment	
	14	40	4/13/24	Total Station Observations for Topography	Traverse
		41	4/15/24	Profile Leveling	
		42	4/17/24	Horizontal Curves	
	15	43	4/20/24	Horizontal Curves	Horizontal Curve Computation, GPS Static Survey
		44	4/22/24	GPS Data Collection	
		45	4/24/24	Vertical Curves	
	16	46	4/27/24	Vertical Curves	Vertical Curve Computation, GPS Kinematic Survey
		47	4/29/24	Public Land Survey System	
		48	5/01/24	Public Land Survey System	
	17	49	5/04/24	**Final Exam: GIS & Surveying**	
		50	5/06/24		
		51	5/08/24		

* Schedule and assignments subject to change. Any changes will be posted in the learning management system